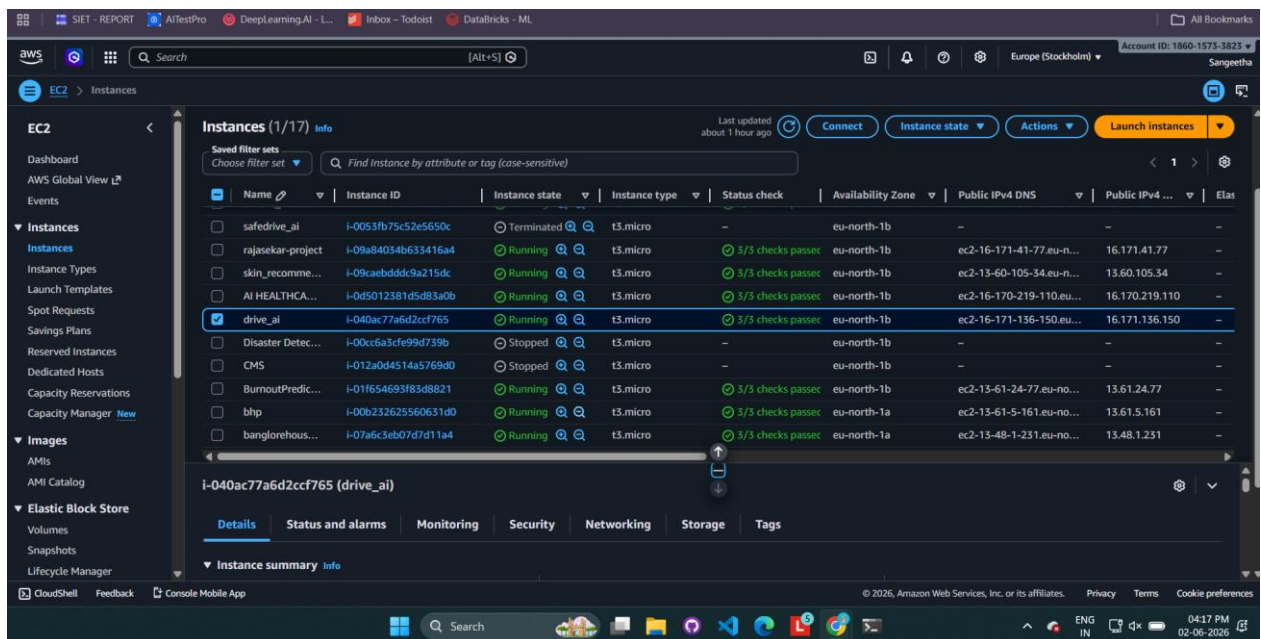


Task – 22 : ML Model Deployment on AWS EC2

This work demonstrates a basic deployment of a machine learning model using Flask on an AWS EC2 instance. A pre-trained ML model along with its vectorizer is integrated into a simple Flask application to perform predictions through a web interface. It also focuses on understanding the fundamentals of cloud deployment, including setting up an EC2 instance, configuring security groups, transferring files from a local system to the server, and running the application on a remote Linux environment. It also involves handling dependency installation and environment setup using Python virtual environments. The main aim is to gain practical exposure to deploying machine learning applications on cloud infrastructure rather than building a full-scale production system.

Steps Performed :

1. Launch an AWS EC2 instance and configure security settings.



2. Connect to the instance using SSH and transfer project files using SCP.

```
PS C:\Users\GOPIKA> scp -i "C:\Users\GOPIKA\Downloads\safedrive.pem" -r "D:\P\ml-app\*" ubuntu@16.171.136.150:/home/ubuntu/ml-app
ubuntu@16.171.136.150: Permission denied (publickey).
PS C:\Users\GOPIKA> scp -i "C:\Users\GOPIKA\Downloads\drive_ai.pem" -r "D:\P\ml-app\*" ubuntu@16.171.136.150:/home/ubuntu/ml-app
app.py          100% 573    2.6KB/s   00:00
model.pkl      100% 9119   42.8KB/s  00:00
requirements.txt 100% 220    1.0KB/s   00:00
mood_check.html 100% 1981   9.3KB/s   00:00
vectorizer.pkl 100% 20KB   93.2KB/s  00:00
PS C:\Users\GOPIKA> ls
```

3. Run the Flask application and access it using the EC2 public IP.

```
(venv) ubuntu@ip-172-31-35-105:~/ml-app$ (venv) ubuntu@ip-172-31-35-105:~/ml-app$
-bash: syntax error near unexpected token `ubuntu@ip-172-31-35-105:~/ml-app$'
(venv) ubuntu@ip-172-31-35-105:~/ml-app$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; vendor preset: enabled)
   Active: active (running) since Tue 2026-06-02 10:28:30 UTC; 1min 28s ago
     Docs: man:nginx(8)
   Main PID: 13971 (nginx)
    Tasks: 3 (limit: 1013)
   Memory: 2.4M (peak: 5.2M)
      CPU: 27ms
   CGroup: /system.slice/nginx.service
           └─13971 "nginx: master process /usr/sbin/nginx"
             └─13973 "nginx: worker process"
               └─13974 "nginx: worker process"

Jun 02 10:28:30 ip-172-31-35-105 systemd[1]: Starting nginx.service: A high performance web server and a reverse proxy...:
Jun 02 10:28:30 ip-172-31-35-105 systemd[1]: Started nginx.service: A high performance web server and a reverse proxy...:

(venv) ubuntu@ip-172-31-35-105:~/ml-app$ sudo nano /etc/nginx/sites-available/mlapp
(venv) ubuntu@ip-172-31-35-105:~/ml-app$ sudo ln -s /etc/nginx/sites-available/mlapp /etc/nginx/sites-enabled
sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
(venv) ubuntu@ip-172-31-35-105:~/ml-app$
```

4. Access the application through the EC2 public IP address

